

Final metric table — all methods, held-out TEST set (seed=42)
Val Loss = validation loss (used for model selection); Test Acc/F1/BalAcc = single held-out evaluation, never seen during training

Part	Method	Val Loss↓	Test Acc↑	Test F1↑	Test BalAcc↑	Wall (s)	GEs	FEs
SL Baseline	SL SGD baseline	0.5382	0.7532	0.5904	0.5561	10.0	2,350	—
Part 1 (RO)	RHC	0.5633	0.7412	0.5181	0.4947	0.0	—	403
Part 1 (RO)	SA	0.5689	0.7438	0.5194	0.5008	0.0	—	801
Part 1 (RO)	GA	0.5696	0.7382	0.5443	0.5308	0.0	—	465
Part 2 (Opt.)	SGD (no mom.)	0.5506	0.7498	0.5663	0.5376	8.0	1,880	—
Part 2 (Opt.)	SGD + mom.	0.5056	0.7827	0.6433	0.6286	8.2	1,880	—
Part 2 (Opt.)	Nesterov	0.5078	0.7700	0.6147	0.5866	8.0	1,880	—
Part 2 (Opt.)	Adam	0.4887	0.7865	0.6567	0.6365	8.8	1,880	—
Part 2 (Opt.)	Adam (no bias-corr.)	0.4888	0.7930	0.6695	0.6338	8.7	1,880	—
Part 2 (Opt.)	Adam ($\beta_1 = 0$)	0.4936	0.7855	0.6658	0.6301	9.0	1,880	—
Part 2 (Opt.)	AdamW	0.4951	0.7745	0.5935	0.5693	8.9	1,880	—
Part 3 (Reg.)	Adam baseline	0.4985	0.7835	0.6445	0.6069	8.8	1,880	—
Part 3 (Reg.)	L2 ($\lambda = 10^{-4}$)	0.4972	0.7785	0.6534	0.6288	8.8	1,880	—
Part 3 (Reg.)	L2 ($\lambda = 10^{-3}$)	0.4975	0.7790	0.5901	0.5457	8.9	1,880	—
Part 3 (Reg.)	Early stopping	0.5011	0.7750	0.6341	0.6124	8.0	1,692	—
Part 3 (Reg.)	Dropout 0.2	0.5228	0.7578	0.5481	0.5344	11.7	1,880	—
Part 3 (Reg.)	Dropout 0.3	0.5433	0.7465	0.5270	0.5185	12.1	1,880	—
Part 3 (Reg.)	Label smooth	0.8409	0.7718	0.5649	0.5325	9.0	1,880	—
Part 3 (Reg.)	Input noise	0.4957	0.7830	0.6369	0.6169	9.5	1,880	—
Part 3 (Reg.)	Best combo	0.8644	0.7600	0.5332	0.5152	12.4	1,880	—